

SCORED CONSISTENCY DISTILLATION OF SD3.5-MEDIUM WITH AN EXACT PHOTOGRAPHIC REWARD

Anonymous authors

Paper under double-blind review

1 METHOD

We distil SD3.5-Medium, a 2.5B-parameter rectified-flow transformer, into a 4-step student. The method has two parts. First, for every training caption the frozen teacher produces several candidate generations, and a scorer that reads a reference photograph of the caption picks one of them; the student is trained only on the picked trajectory. Second, an exact photographic reward term pulls the student’s own predictions toward the same photograph during training. Both parts use the same underlying quantity, the DINOv2 similarity of an image to the photograph the caption describes, and neither reads the caption text or any language model.

1.1 PRELIMINARIES: RECTIFIED-FLOW DISTILLATION

The teacher is a rectified-flow model with frozen weights θ_T . A clean latent x_0 and Gaussian noise $\varepsilon \sim \mathcal{N}(0, I)$ define a straight path $z_\sigma = (1 - \sigma)x_0 + \sigma\varepsilon$, $\sigma \in [0, 1]$, and the network predicts the velocity $v = \varepsilon - x_0$ conditioned on the caption c . Sampling is Euler integration on a σ grid, $z_{k+1} = z_k + (\sigma_{k+1} - \sigma_k) v_\theta(z_k, \sigma_k, c)$, with classifier-free guidance $v^{\text{cfg}} = v_\theta(z, \sigma, \emptyset) + w(v_\theta(z, \sigma, c) - v_\theta(z, \sigma, \emptyset))$ at $w = 7$ for the teacher. The student v_θ is a full copy of the same transformer, initialised from θ_T , and is the only network that trains; the teacher, the text encoders and the VAE decoder Dec stay frozen throughout. There is no momentum target, no exponential moving average and no stop-gradient copy of the student anywhere in the recipe: every training target below is a deterministic function of the frozen teacher.

Consistency distillation trains the student to jump directly to the clean-latent estimate that the teacher reaches only after several further steps, so that a 4-step Euler rollout of the student reproduces an 8-step teacher trajectory. This is the sole mechanism that turns an 8-step teacher into a 4-step student; nothing below changes it. What varies is which trajectory the student is trained on, and what, besides matching that trajectory, the training loss asks the student’s own predictions to do.

1.2 CANDIDATE GENERATION AND PHOTOGRAPHIC SCORING

For every training caption c with reference photograph x^{ref} (its COCO source image), we draw $N = 4$ candidate teacher trajectories from four independent seeds and score each one against the photograph.

Candidates. Candidate j uses Gaussian noise $\varepsilon^{(j)}$ seeded deterministically from the caption index, and is an 8-step Euler rollout of the teacher at guidance $w = 7$:

$$\varepsilon^{(j)} \sim \mathcal{N}(0, I) \text{ (seeded by caption and } j), \quad z_0^{(j)}, \dots, z_8^{(j)} = \text{Euler}_{K=8}(\varepsilon^{(j)}; v_T^{\text{cfg}}, c). \quad (1)$$

Only the seed, not the trajectory, needs to be stored: any candidate can be re-rolled exactly from its seed whenever it is needed again, at scoring time and later at training time.

Scoring. Each candidate is decoded, $I_j = \text{Dec}(z_8^{(j)})$, and embedded with a frozen DINOv2-B vision transformer. We drop the CLS token, mean-pool the $P=256$ patch tokens and ℓ_2 -normalise:

$$u(I) = \frac{\frac{1}{P} \sum_{p=1}^P h_p(I)}{\left\| \frac{1}{P} \sum_{p=1}^P h_p(I) \right\|}, \quad S_j = \langle u(I_j), u(x^{\text{ref}}) \rangle. \quad (2)$$

$u(x^{\text{ref}})$ is computed once per caption. This scorer needs no caption text, no VLM and no learned preference model: it is a fixed, off-the-shelf self-supervised image encoder, applied identically to a generated candidate and to a real photograph.

Selection. We select the best candidate by hard argmax over the score,

$$j^* = \arg \max_{j \in \{1, \dots, N\}} S_j, \quad (3)$$

and every subsequent training visit of caption c trains only on trajectory j^* . Selection is computed once, offline, before training starts, and is never revisited: no gradient flows through the scorer, and the training loop simply re-rolls trajectory j^* from its stored seed. A random baseline replaces j^* with a uniform draw over $\{1, \dots, N\}$, stored so that every training seed sees the same draw; this isolates the effect of *which* trajectory a caption trains on from everything else in the recipe.

1.3 CONSISTENCY DISTILLATION LOSS

Let $z_0, \dots, z_8$ be the selected teacher trajectory. Training supervises a window of states $\mathcal{K} = \{3, 4, 5, 6, 7\}$ on the 8-step grid. For $k \in \mathcal{K}$, the teacher’s realised chord across its own step and the clean-latent estimate it implies are

$$\bar{v}_k = \frac{z_{k+1} - z_k}{\sigma_{k+1} - \sigma_k}, \quad \hat{x}_0^{(k)} = z_k - \sigma_k \bar{v}_k, \quad (4)$$

which is exact algebra for a straight path. The student is placed at a *noisier* state than the one the target was read from: for each k we draw $\Delta_k \sim \mathcal{U}\{1, 2, 3\}$ independently and afresh on every visit, set $s = k - \Delta_k$, and the student forms its own clean-latent estimate with a single conditional forward pass and no guidance,

$$\hat{x}_0^{(s)} = z_s - \sigma_s v_\theta(z_s, \sigma_s, c). \quad (5)$$

The consistency loss is a pseudo-Huber distance in x_0 -space, averaged over the window:

$$\ell_c(\theta) = \frac{1}{|\mathcal{K}|} \sum_{k \in \mathcal{K}} \rho(\hat{x}_0^{(k-\Delta_k)} - \text{sg}[\hat{x}_0^{(k)}]), \quad \rho(u) = \sqrt{\|u\|_2^2 + c_h^2} - c_h, \quad (6)$$

with $\text{sg}[\cdot]$ a stop-gradient and c_h a small constant scaled to the latent dimensionality. Because the student is trained with a single conditional pass against $w=7$ teacher trajectories, guidance is absorbed into its prediction: the student is sampled at $w=1$, with no classifier-free guidance at inference.

1.4 EXACT PHOTOGRAPHIC REWARD

Selection alone only changes which of the teacher’s four trajectories the student is trained on; it never moves the student’s own predictions toward the photograph directly. The reward does. On every training update, in addition to matching the (stop-graduated) teacher targets, the student’s own clean-latent estimates at the two least-noisy supervised states are decoded, re-embedded with the same DINOv2 scorer used for selection, and pulled toward the reference photograph by gradient:

$$r_c(\theta) = \frac{1}{2} \sum_{i=1}^2 \langle u(\text{Dec}(\hat{x}_0^{(s_i)})), u(x^{\text{ref}}) \rangle, \quad (7)$$

where Dec , DINOv2 and $u(x^{\text{ref}})$ are exactly the frozen decoder and scorer of Section 1.2: the reward is differentiated through the VAE decoder and through DINOv2, both with frozen weights, into the student. No stop-gradient is used inside r_c . The full per-caption training objective is

$$\mathcal{L}(\theta) = \mathbb{E}_c[\ell_c(\theta)] - \lambda \mathbb{E}_c[r_c(\theta)], \quad (8)$$

combining Equations 6 and 7. The weight $\lambda=15.504$ is set once, before training, so that the raw reward gradient’s norm is 20% of the raw consistency gradient’s norm, measured over a short probe at the start of training; the realised ratio drifts to 25–40% over a full run. The method reads the reference photograph twice: once, offline, to choose which of four trajectories to train on, and once, inside every training step, to pull the student’s own predictions toward it directly.

Algorithm 1 Scored consistency distillation with an exact photographic reward

Phase 1 – Choose a trajectory for every caption (once, before training).

- 1: **for** each training caption c , with reference photograph x^{ref} **do**
- 2: Generate $N=4$ candidate images by rolling out the frozen teacher from 4 seeds ▷
- 3: Section 1.2, first paragraph
- 3: Score every candidate’s similarity to x^{ref} with the DINOv2 scorer $u(\cdot)$ ▷ Equation 2
- 4: Keep only the seed of the best-scoring candidate j^* ▷ Equation 3; nothing else is stored
- 5: **end for**

Phase 2 – Train the student (repeat every optimiser step).

 - 6: **for** each optimiser step **do**
 - 7: Pick a caption c ; regenerate its kept trajectory j^* from the stored seed
 - 8: At 5 noise levels along that trajectory, compute the teacher’s target and the student’s (one-step-noisier) prediction ▷
 - 9: Section 1.3
 - 9: $\ell_c \leftarrow$ average distance between the student’s predictions and the teacher’s targets ▷
 - 10: Equation 6
 - 10: $r_c \leftarrow$ average photographic reward of the student’s 2 least-noisy predictions ▷ Equation 7
 - 11: Take a gradient step on $\ell_c - \lambda r_c$ (clip the gradient norm to 1; AdamW) ▷ Equation 8
 - 12: **end for**

1.5 TRAINING ALGORITHM

Algorithm 1 states the full method in two phases: candidate generation and selection happen once, before training starts, for every caption; the training loop then repeats one simple update. Every quantity below was defined precisely in Sections 1.2–1.4 above; the pseudocode intentionally does not re-derive the equations, only names which one each line uses, so that the flow of the method stays visible at a glance.

At inference the student is sampled with 4 Euler steps on its own grid, $w=1$, with no guidance and no candidate generation: Phase 1 and the reward only apply at training time. The trained student is a single frozen network; nothing about candidate pools, scorers or the reward survives into the deployed model.

2 EXPERIMENTAL SETUP

Data. Training captions come from COCO `train2017`, each paired with the photograph it was originally written for; that photograph is used only by the scorer and the reward, and never enters the student’s training input. Two pools are used: a **3k pool** of 3,000 captions drawn once with a fixed seed, used for every multi-seed comparison, and a **118k pool** of 113,948 captions, used to test whether the recipe holds at scale. All evaluation uses a separate, sealed pool built once with a fixed seed; evaluation references and fidelity targets come from COCO `val2017`, so no evaluation caption or photograph is seen in training.

Training configurations. Table 1 gives both configurations. Both use AdamW, global gradient-norm clipping at 1, fp32 master weights under bf16 autocast, and a frozen teacher, text encoders and VAE. The 118k configuration uses 4 GPUs in data parallel, one caption per GPU per step; its reported student is the uniform weight average of the last five saved checkpoints, which is necessary because consecutive checkpoints otherwise differ by up to 0.035 CompBench under this schedule.

Evaluation. We report alignment with T2I-CompBench (8 categories: color, shape, texture, 2D-spatial, 3D-spatial, numeracy, non-spatial, complex; one image per prompt unless stated, and the official ten-images-per-prompt protocol where noted) and GenEval2 (object, count, attribute, position and verb skills, atom counts 3 to 10, scored as the geometric mean over atoms). We report fidelity with FID, CMMD, precision and recall against 5,000 held-out COCO `val2017` photographs. Every arm and every seed is evaluated on exactly the same evaluation pool. Unless stated, all reported numbers are for a student sampled with 4 Euler steps, $w=1$.

Table 1: Training configurations. Both use identical code, differing only in scale.

	3k configuration	118k configuration
Captions	3,000	113,948
GPUs	1	4, data-parallel, one caption per GPU per step
Updates	6,000 (2 passes)	56,974 (2 passes)
Training seeds	3 – 5	3
Learning rate	10^{-5}	2×10^{-5}
Warm-up	300 steps	1,000 steps
Reward weight λ	15.504	– (Section 3.2)
Reported weights	final step	uniform average, last 5 checkpoints

Table 2: Alignment on the 3k pool, five training seeds, averaged checkpoints, one image per prompt. Deltas are seed-paired against the row above, mean $\pm$ standard deviation over seeds.

Recipe	CompBench	Δ	GenEval2 ($\times 100$)	Δ
Random selection, no reward (baseline)	0.4751 ± 0.0032	–	22.47 ± 0.68	–
Argmax selection, no reward	0.4868 ± 0.0031	$+0.0117 \pm 0.0058$ ($p=0.011$)	23.92 ± 1.29	$+1.46 \pm 1.38$
Argmax selection + exact reward (ours)	0.4923 ± 0.0021	$+0.0055 \pm 0.0021$ ($p=0.004$)	24.41 ± 0.77	$+0.49 \pm 1.30$
Ours vs. baseline	$+0.0172 \pm 0.0051$	CompBench ($p=0.002$),	$+1.94 \pm 0.68$	GenEval2 ($p=0.003$)

3 RESULTS

3.1 ALIGNMENT

Table 2 compares three training recipes on the 3k pool, matched in every other respect: random trajectory selection with no reward (the naive baseline), argmax selection with no reward, and argmax selection with the exact photographic reward (our full method). Both components help, and they combine additively: selection alone is worth $+0.012$ CompBench over the baseline, the reward alone (on top of selection) adds a further $+0.0055$ ($p = 0.004$), and the two together are $+0.0172$ CompBench ($p = 0.002$) and $+1.94$ GenEval2 ($p = 0.003$) over the naive baseline, with every seed positive on both benchmarks.

A one-factor sweep around this recipe (Appendix, Section 4) finds that its reward weight is conservative: raising λ to 31 and rewarding all five supervised states instead of the two least-noisy reaches **0.5082** CompBench, our strongest configuration on this metric, $+0.0153$ over the recipe above ($p = 0.012$); this configuration trades some of its gain on GenEval2 and FID for its CompBench and CMMD gain (Appendix), so Table 2’s recipe is the one we report throughout the rest of this section.

Per-category breakdown. Table 3 breaks the CompBench gain down by category. The gain is not concentrated in one easy category: every one of the eight categories moves in the recipe’s favour, and the two spatial-relation categories, historically the hardest for text-to-image models, show the largest absolute gains.

GenEval2 breakdown. Table 4 breaks GenEval2 down by skill and by the number of atomic sub-conditions a prompt has (3 to 10). Attribute binding and position, the two skills closest to CompBench’s color and spatial categories, gain the most; the gain persists across every atom-count bucket, i.e. it does not come only from the easiest, lowest-atom prompts.

3.2 SCALING TO 118,000 CAPTIONS

Selection was re-tested at 118k captions with three training seeds (the exact reward has so far only been validated at the 3k scale above). Table 5 gives the result: $+0.0175$ CompBench ($p = 10^{-14}$) and $+1.29$ GenEval2 over the naive baseline. Under the official ten-images-per-prompt CompBench protocol, the far less noisy evaluation, the gain is $+0.0131$ at $p = 2 \times 10^{-41}$ (Table 6), and it holds in every one of the eight categories under both protocols. The scaled-up student closes most of the gap to its own 28-step teacher: the averaged patch student is 0.021 CompBench below the 28-step teacher and 0.030 above its 8-step source.

Table 3: CompBench by category, 3k pool, five seeds, one image per prompt, averaged checkpoints.

Category	Random, no reward	Ours	Δ (seed-paired)
Color	0.8086 ± 0.0077	0.8228 ± 0.0110	$+0.0142 \pm 0.0140$ ($p=0.086$)
Shape	0.5517 ± 0.0139	0.5672 ± 0.0059	$+0.0155 \pm 0.0143$ ($p=0.071$)
Texture	0.7151 ± 0.0084	0.7311 ± 0.0048	$+0.0160 \pm 0.0125$ ($p=0.046$)
2D-spatial	0.2110 ± 0.0196	0.2459 ± 0.0115	$+0.0349 \pm 0.0231$ ($p=0.028$)
3D-spatial	0.3030 ± 0.0058	0.3234 ± 0.0134	$+0.0204 \pm 0.0126$ ($p=0.022$)
Numeracy	0.5313 ± 0.0113	0.5591 ± 0.0135	$+0.0278 \pm 0.0224$ ($p=0.050$)
Non-spatial	0.3104 ± 0.0004	0.3116 ± 0.0013	$+0.0013 \pm 0.0012$ ($p=0.072$)
Complex	0.3700 ± 0.0027	0.3774 ± 0.0036	$+0.0074 \pm 0.0048$ ($p=0.026$)
Mean	0.4751 ± 0.0032	0.4923 ± 0.0021	$+0.0172 \pm 0.0051$ ($p=0.002$)

Table 4: GenEval2, 3k pool, five seeds, averaged checkpoints. Overall is the official geometric mean over atoms; atom mean is the arithmetic mean; collapsed is the fraction of prompts with a geometric-mean score under 0.05, i.e. at least one atom entirely wrong.

Measure	Random, no reward	Ours	Δ
Overall (official)	22.5 ± 0.7	24.4 ± 0.8	$+1.9 \pm 0.7$ ($p=0.003$)
Atom mean (arithmetic)	61.9 ± 0.8	64.0 ± 0.6	$+2.2 \pm 1.1$ ($p=0.010$)
Collapsed prompts (%)	36.5 ± 3.6	31.4 ± 1.5	-5.2 ± 4.2
Skill: object	83.7 ± 1.5	85.9 ± 0.7	$+2.3 \pm 2.0$
Skill: count	35.9 ± 1.1	37.5 ± 1.2	$+1.6 \pm 1.9$
Skill: attribute	70.8 ± 1.2	73.5 ± 1.0	$+2.7 \pm 1.9$ ($p=0.034$)
Skill: position	48.3 ± 2.9	51.8 ± 0.8	$+3.5 \pm 3.0$
Skill: verb	23.5 ± 4.6	26.5 ± 2.2	$+3.0 \pm 4.8$
Atoms = 3	44.2 ± 1.8	46.2 ± 1.5	$+2.0 \pm 1.7$
Atoms = 4	34.7 ± 2.7	37.0 ± 2.9	$+2.3 \pm 4.9$
Atoms = 5	24.9 ± 0.4	27.2 ± 3.2	$+2.3 \pm 3.5$
Atoms = 6	22.0 ± 1.1	22.6 ± 1.0	$+0.6 \pm 1.6$
Atoms = 7	17.8 ± 1.7	18.5 ± 2.2	$+0.7 \pm 3.8$
Atoms = 8	13.6 ± 1.1	16.1 ± 1.4	$+2.6 \pm 2.0$ ($p=0.046$)
Atoms = 9	13.1 ± 1.0	16.8 ± 1.0	$+3.8 \pm 1.4$ ($p=0.004$)
Atoms = 10	9.5 ± 1.3	10.8 ± 1.1	$+1.4 \pm 2.2$

3.3 FIDELITY

Selection and the reward do not trade alignment for image quality; both improve fidelity to real photographs at the same time. Table 7 gives 3k-pool fidelity for the three recipes of Table 2; the reward is the larger lever here, lowering CMMD by 0.07–0.10 with every seed improving. Table 8 gives the 118k-scale result for selection: CMMD, precision and recall all improve on every one of the three seeds, and FID is unchanged within its split-half noise.

3.4 QUALITATIVE RESULTS

Table 3’s 2D-spatial category is where the recipe gains the most, and Figure 1 shows two concrete T2I-CompBench 2D-spatial prompts behind that number. Each prompt names two objects and a specific left/right or top/bottom relation between them; we generated our student at its trained 4-step budget and, to check the placement was not a 4-step accident, again at 8 and 28 steps, and compared against the frozen 28-step teacher sampled with classifier-free guidance ($w=7$), all four images per prompt from identical initial noise. In both prompts every one of our three student samples places the two named objects in the relation the prompt asks for; the guided teacher does not, placing them the other way around (“a boy on the left of a balloon”) or without the second object resting on the first (“a candle on the top of a chicken”). These two prompts were found by inspecting images, not by scores alone: several other high-scoring-gap prompts we checked turned out to be

Table 5: 118k captions, 56,974 updates, three training seeds, weight-averaged students.

Recipe	CompBench (mean, Δ , p)	GenEval2 (mean, Δ , p)
Random selection (baseline)	0.4668	20.29
Argmax selection (ours)	0.4843, +0.0175, 10^{-14}	21.58, +1.29, 0.02

Table 6: Official ten-images-per-prompt T2I-CompBench, 118k weight-averaged students, three seeds.

Recipe	CompBench (mean, Δ , p)
Random selection (baseline)	0.4719
Argmax selection (ours)	0.4850, +0.0131, 2×10^{-41}

genuinely correct in both the student’s and the teacher’s image, with the automatic evaluator’s score gap coming from image framing rather than a visible placement error, and are not shown here. At its trained 4-step budget the student’s automatic CompBench score is 0.97 against the teacher’s 0.00 on the first prompt and 0.89 against 0.00 on the second; we did not re-run the automatic evaluator on the 8-step and 28-step student images, so only the 4-step column carries a formal score. The broader point is qualitative: a distilled few-step student is not, skill by skill, a strictly degraded copy of its many-step guided teacher.

4 ADDITIONAL ABLATIONS

The following ablations were run on the 3k pool and are reported in full in the appendix; only their titles are given here.

- Selection rule: argmax vs. Boltzmann-weighted vs. sampled vs. uniform candidate weighting.
- Reward weight λ and number of rewarded states: a one-factor sweep and their combination.
- Which supervised states are rewarded: least-noisy vs. noisiest.
- A decode-free latent scorer and a learned latent reward, as alternatives to decoding.
- Consistency-target horizon: 1-step Tweedie estimate vs. a 2-step chord vs. the trajectory endpoint.
- Training objective: x_0 regression vs. segment-velocity imitation.
- Teacher quality: candidate generation and training with a 16-step and a 28-step teacher.
- Regression onto the reference photograph as a second training target.
- An evolving (exponential-moving-average) teacher in place of the frozen teacher.
- Number of candidates ($N=4$ vs. $N=8$) and optimiser batch size, at the 118k scale.
- Inference step count: sampling the 4-step student at 2, 8, 16 and 28 steps.

Table 7: Fidelity, 3k pool, five seeds, averaged checkpoints, 5,000 COCO val2017 captions.

Recipe	CMMD ↓	Precision ↑	Recall ↑	FID ↓
Random selection, no reward	0.846 ± 0.035	0.478	0.055	–
Argmax selection, no reward	0.794 ± 0.023	0.497	0.066	–
Argmax selection + exact reward (ours)	0.726 ± 0.017	0.522	0.082	–

Table 8: Fidelity at 118k, weight-averaged students, three seeds (s0 / s1 / s2), 5,000 COCO val2017 captions. The 28-step teacher is shown for reference, not as a fair comparison at 4 steps.

Recipe	FID ↓	CMMD ↓	Precision ↑	Recall ↑
Random (baseline)	31.08 / 30.97 / 31.44	0.84 / 0.84 / 0.83	0.485 / 0.489 / 0.503	0.059 / 0.066 / 0.062
Argmax selection (ours)	30.98 / 30.83 / 31.47	0.78 / 0.79 / 0.78	0.522 / 0.511 / 0.522	0.071 / 0.068 / 0.063
Teacher, 28 steps, $w=7$	29.21	0.64	0.549	0.156



Figure 1: Two T2I-CompBench 2D-spatial prompts, identical initial noise per prompt (columns are not independently seeded). Our student (argmax selection + exact reward, no guidance) places both named objects in the requested relation at 4, 8 and 28 sampling steps alike; the frozen teacher with guidance ($w=7$) does not, at 28 steps. Green borders: the requested relation is satisfied. Red border: it is not. Automatic CompBench 2D-spatial scores (only computed for the trained 4-step student and the teacher): 0.97 vs. 0.00 (top row), 0.89 vs. 0.00 (bottom row).